

ai-os-new

Race / Lifecycle Re-Audit Report

Verification audit after bug fixes. Original: 15 findings. Remaining: 15 findings.

Severity	Count
CRITICAL	0
HIGH	3
MEDIUM	9
LOW	3
Total	15

RC -
01

MEDIUM

VoiceButton: setState after unmount from SpeechRecognition callbacks

Files: VoiceButton.tsx

Problem: recognition.onend/onerror call setIsListening(false) with no mounted guard or cleanup.

Fix: Add cleanup useEffect that stops recognition and nulls callbacks.

RC -
02

MEDIUM

DebateWorkspacePanel: setTimeout polling chain not cleaned up

Files: DebateWorkspacePanel.tsx

Problem: setTimeout polling chain (500ms, up to 20 attempts). No cleanup. setState on unmounted component.

Fix: Track timeout ID and clearTimeout in useEffect return.

RC -
03

HIGH

HealthService.checkKey: Timeout timer leaked when adapter wins race

Files: health-service.ts

Problem: Promise.race between adapter.checkHealth and timeout setTimeout. When adapter wins, timeout is never cleared - fires after 15s.

Fix: Capture timeout ID and clearTimeout after race resolves.

RC -
04

MEDIUM

CrossTabStateSync: localStorage storage event listener never removed

Files: cross-tab-state.ts

Problem: initLocalStorageFallback adds storage event listener but destroy() never removes it.

Fix: Store handler reference and removeEventListener in destroy().

RC -
05

MEDIUM

CircuitBreaker: inFlightHalfOpen counter can go negative across reset cycles

Files: circuit-breaker.ts

Problem: reset() doesn't zero inFlightHalfOpen. Stale decrements from previous half-open cycle can make counter negative in new cycle.

Fix: Zero inFlightHalfOpen in reset(). Use Math.max(0, counter-1) in finally.

RC -
06

HIGH

ProxyHealthMonitor: concurrent performCheck corrupts consecutiveFailures

Files: proxy-health-monitor.ts

Problem: No guard against concurrent performCheck calls. scheduleCheck runs immediately AND on interval. checkNow can overlap. Both read/mutate same status object.

Fix: Add per-route in-flight guard (Set of checking routes).

RC -
07

MEDIUM

debounce/throttle utilities lack cancel() method

Files: debounce.ts

Problem: debounce returns function with no .cancel(). Pending timers fire after component/service destroyed. throttle drops all calls during window.

Fix: Add cancel() method to debounce. Add trailing edge option to throttle.

RC -
08

HIGH

RaceExecutor: Promise.race silently drops slower successes

Files: race-executor.ts

Problem: When fastest candidate resolves, race completes. Slower potentially-better responses are aborted and lost. Only captures pre-winner failures.

Fix: Use Promise.allSettled with timeout instead of Promise.race. Or at minimum track overtaken candidates.

RC -
09

LOW

ChatService: activeRequests Map entry leaks on edge case

Files: chat-service.ts

Problem: activeRequests.set() happens before try block. If code between set and try throws, finally won't clean up the entry.

Fix: Move activeRequests.set inside try block.

RC - 10	MEDIUM	DebateService: isExecutingRound set too late
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Files: debate-service.ts

Problem: isExecutingRound=true is set AFTER getNextParticipant() await. During that await, another timer can see isExecutingRound=false and start concurrent round.

Fix: Set isExecutingRound=true BEFORE getNextParticipant() call.

RC - 11	LOW	SchedulerService: same schedule triggered twice during slow runSchedule
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Files: scheduler-service.ts

Problem: checkSchedules awaits runSchedule sequentially. During await, timer can fire again and re-trigger same schedule (not yet updated).

Fix: Add isChecked guard.

RC - 12	MEDIUM	ResumableStream: switchProvider mutates state before confirming new provider works
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Files: resumable-stream.ts

Problem: switchProvider sets state.provider=newProvider and state.status=active before resume(). If resume fails, state is inconsistent with no rollback.

Fix: Save previous state and rollback on failure.

RC - 13	MEDIUM	HealthPanel: probe button setState without mounted guard
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Files: HealthPanel.tsx

Problem: Quick Test All button starts async probeAll() and calls setProbeResults/setProbeLoading without isMountedRef check.

Fix: Wrap post-await setState with isMountedRef.current check.

RC - 14	LOW	DebateRuntimeAdapter: double-finalize on stop during startSession
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Files: debate-runtime-adapter.ts

Problem: startDebate is fire-and-forget. If stop() called during startSession, finalize() called twice - once by stop, once by promise chain.

Fix: Guard with sessionId === runtimeId check.

RC - 15	MEDIUM	key-storage-hydrator: TOCTOU on retry after failure
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Files: key-storage-hydrator.ts

Problem: _hydrationPromise cleared in finally, allowing re-entry. Between finally clearing and .then processing, a new hydration could start.

Fix: Don't clear _hydrationPromise in finally. Only clear on explicit reset or failure.